

MULTIMODEL DARK PATTERN DETECTION SYSTEM

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ABSTRACT

Dark patterns are deceptive user interfaces designed to guide users towards actions that are neither intended nor expected, such as forced subscriptions or artificial urgency. Currently, automatic detection methods are only based on reading texts and ignore visual cues that could have important impact on user habits. We introduce a lightweight multimodal dark pattern detection framework, that involves both textual machine-learning and heuristic-based visual feature extraction. The system integrates TF-IDF and Multinomial Naive Bayes classifier with browser-level analysis of its visual attributes including color, positioning, size, and contrast. A weighted fusion mechanism combines textual and visual scores to create final predictions. We conduct the experimental test on annotated e-commerce data which shows a 6–11% improvement in detection performance over text-only methods whilst also producing low computational overhead, and requiring no GPU support. They emphasize the importance of integrating visual and linguistic signals to create reliable and wide-reaching dark pattern detection.

Index Terms— Dark patterns, Deceptive user interfaces, Multimodal detection, Visual-textual fusion, Human-computer interaction, Usable security, Privacy protection, Naive Bayes classification, Browser-based analysis, E-commerce systems.

1. INTRODUCTION

Digital platforms have shown an increasing reliance on interfaces that design with user behavior in mind. Many design practices focus on usability improvement and user engagement, but some intentionally take advantage of cognitive biases to manipulate users towards decisions or actions they would not otherwise make. These are deceptive practices in face, frequently known as dark patterns, they include the use of artificial urgency, false button hierarchies, undisclosed subscription plans and obstructive cancellation flows. These are

the patterns that are very common in e-commerce where small manipulation can greatly change purchase intentions and user consents.

Prior research has investigated dark pattern detection using manual audits, rule-based systems and text-based machine-learning methods. While useful for generating valuable insights, however, many of these automated solutions are centered around linguistic cues. However, user interfaces rarely communicate deception in text alone. Color intensity, button size, spatial orientation, and contrast are visual details that significantly affect the user’s perceptual context and engagement during interactions. For instance, a bright colored “Subscribe” button overlaid above the fold with a visually muted “Cancel” option can direct a user’s intent, independent of the textual content. The over-reliance on only text detection processes makes ignoring these visual signals inefficient.

This work responds to this limitation by introducing one-stop dark pattern detection method, which combines the concepts of extrapolating classification and visual feature analysis. The system incorporates a weighted fusion mechanism to capture their complementary relationship, so that visual and linguistic signals are integrated in a complementary manner, without treating visual and linguistic signals in isolation. The textual component exploits TF-IDF vectorization with a Multinomial Naive Bayes classifier, whereas the visual component retrieves browser-level functionalities (e.g., colors associated with urgency, levels of element prominence, contrast ratios and spatial positioning) directly from the Document Object Model (DOM).

One of the main design aims of this work is usability. The proposed framework is lightweight and can be set-up in practice, unlike existing learning-based frameworks that necessitate considerable computational resources. The system is run as a Chrome browser extension with Flask-based backend without GPU processing. This design allows real-time detection with low latency while retaining interpretability based on

confidence scores and feature-based reasoning.

Experimental testing using annotated e-commerce dark pattern data shows integrated visual features improves detection performance in comparison to text-only baselines. The implications suggest that visual indications add meaningful supplementary features, especially in reducing false negatives where textual inputs cannot provide enough.

Applying this system of combining text and visual indicators of manipulation, this work offers a practical and computationally efficient approach to dark pattern detection. Our framework enables the development of user-centric tools to boost transparency and support participatory decision making for informed users, thus, driving forward automated detection mechanisms.

2. LITERATURE REVIEW

Dark patterns were first described as deceptive interface designs designed to manipulate user decisions through deceptive interaction strategies. Brignull developed the term and described common manipulative design practices present within online systems [1]. This seminal work provided a theoretical underpinnings for the following academic research. Based on this premise, Gray et al. formalized a structured taxonomy of dark patterns and categorized deceptive strategies: obstruction, misdirection, forced action, sneaking [2]. Their work established a structured classification framework that allowed for large empirically-oriented analysis on large scale. Building on this aspect of our work, Mathur et al. conducted a comprehensive crawl of 11K+ shopping websites and highlighted the global spread of dark patterns and the ability to manipulate them in e-commerce environments [3]. The study demonstrated the problems posed by manually identifying deceptive interfaces at scale. In addition to prevalence studies, behavioral and regulatory issues have been studied in addition. Luguri Strahilevitz experimentally show interface manipulations play a huge role in user decision making, showing that deception frequently lies in presentation and structure rather than the language of text [4]. Due to their computational efficiency, text-based machine learning methods have widely established the application. Logistic regression with bag-of-words representations studies proved that urgency phrases and scarcity indicators can be successfully recognized utilizing statistical models [5]. Similarly, Naive Bayes classifiers with TF-IDF vectorization show competitive performances in detecting deceptive textual patterns with low computational complexity [6]. Advanced transformer-based architectures such as BERT increased the contextual understanding as well as classification accuracy at vastly higher computational cost [7]. Awofisayo and Moran employed different object detection methods to annotated screenshots with the UI, showing that there were deceptive design elements which are detected based on the spatial layout and visual prominence analysis [8]. Jang et al. applied

this approach in real-time object-detection frameworks and observed a better localisation for visual manipulative components in the interface scenarios [9]. Though effective, these approaches are based on heavily annotated visual datasets and GPU-dependent deep learning architectures. More recently, multimodal systems have also started a textual and visual integrative analysis. Mansur et al. showed a paradigm merging natural language processing with screenshot-based visual recognition [10], highlighting the complementary role of linguistic and visual signals in dark pattern detection. Nevertheless, current multimodal approaches typically rely heavily on computationally intensive deep learning pipelines and do not have lightweight deployment mechanisms. From the previous literature, I may identify three key gaps. First, the text-only methods fall short of capturing manipulation conveyed through visual emphasis. Second, vision-based deep learning methods are computationally expensive. Third, lightweight multimodal systems which operate directly from browser DOM are restricted. These observations drive the strategy, which integrates text-based classification with heuristic visual feature extraction for better detection performance while still maintaining on-the-fly deployability.

3. PROPOSED METHODOLOGY

3.1. System Overview

The solution proposed in this paper is a lightweight multimodal model that aims to detect dark patterns in web interfaces through text classification and visual analysis of interface content. Based on client/server we have a frontend and backend that is provided by a Chrome extension as well as a backend server built on Flask. Web browser extension extracts visible textual content and associated visual attributes straight from Document Object Model (DOM). The acquired data are sent to the backend for probabilistic text classification and visual scoring. A fusion system combines the two modalities for final deception score. Detected elements are emphasized in real time within the user's browser.

3.2. Text Extraction Module

The text extraction module works through the DOM itself. Browser-native APIs parse all visible user-facing text elements—buttons, banners, promotional messages, pop-ups, etc. Such as buttons on the app browser, which are visible to the naked eye, make it obvious what the user is typing. We reject hidden, script-generated, and non-visible elements to remove noise. The extracted textual segment is pre-processed with normalization, tokenization, stop-word removal and vectorization by Term Frequency–Inverse Document Frequency (TF–IDF). This characterization retains contextual information and computational efficiency.

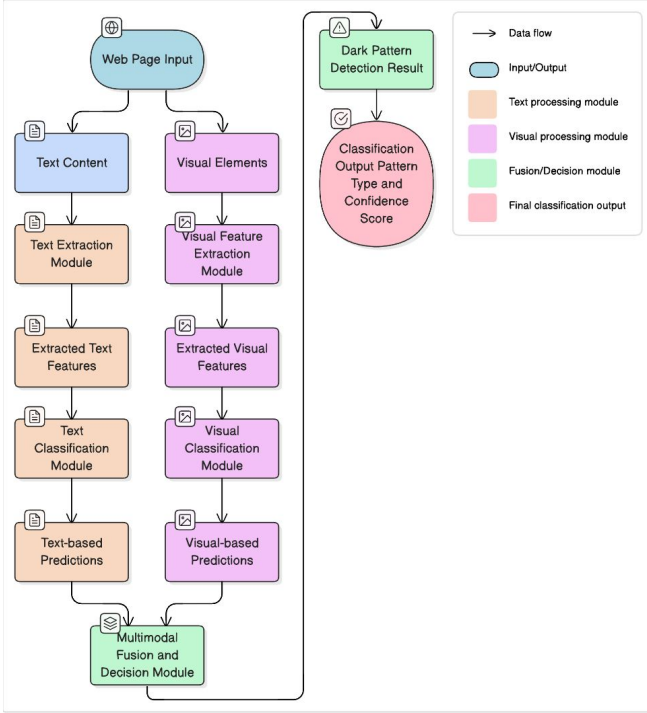


Fig. 1. The complete pipeline is presented as block representation depicted in Fig. 1.

3.3. Visual Feature Extraction Module

The system extracts visual features together with textual signals when considering deceptive interface design. A normalized set of visual features for each DOM element is derived utilizing `getComputedStyle()` and viewport metrics.

Table 1. Visual Features Used in Deception Analysis

Feature	Description
Urgency Color Score	Detects red/orange dominance
Button Prominence	Measures solid vs outline styling
Font Size Ratio	Relative emphasis versus surrounding text
Above-the-Fold	Visible in initial viewport
Fixed Position	Sticky banners or pop-ups
Contrast Ratio	Foreground-background visibility
Scarcity Indicators	Words like “Only”, “Limited”

3.4. Text Classification Module

Text classification is performed using a Multinomial Naive Bayes model trained on 1,825 annotated dark pattern instances spanning seven commonly observed categories. The classifier estimates the posterior probability of each text segment belonging to a deceptive class.

For a given text segment x_i , the model computes:

$$P(y | x_i) \propto P(y) \prod_{k=1}^n P(w_k | y)$$

where w_k represents tokens in the segment and y denotes the class label.

The text-only baseline achieves an accuracy of 87%, which means that although language cues bring strong performance potential, they are insufficient for visually-centred deceptive purposes.

3.5. Multimodal Fusion and Decision Strategy

The fused score is computed as:

$$\text{Score}_{\text{fused}}(i) = w_T T_i + w_V V_i$$

where:

T_i is the text classifier probability,

V_i is the aggregate visual deception score,

$w_T = 0.7$ and $w_V = 0.3$ are the fusion weights.

An element is deceptive if:

$$\text{Score}_{\text{fused}}(i) > \theta$$

where $\theta = 0.5$ is empirically determined.

This multimodal approach increases the overall detectability to 92% and recall rates to be higher than the text-only baseline..

3.6. Computational Efficiency

The framework is lightweight and CPU-efficient. Basic e-commerce sites run DOM parsing and feature extraction at reduced latency (<500 ms). Backend inference is less than 50 ms per request. The system does not need GPU acceleration or deep neural architecture, making it friendly for real-time deployment in a consumer’s browser.

4. RESULTS AND DISCUSSION

4.1. Experimental Setup

On a dataset with 1,825 annotated dark pattern instances of seven categories the proposed framework was tested. A train-test split of 80:20 was applied. The text classifier employed TF-IDF vectorization with a Multinomial Naive Bayes model. Accuracy, Precision, Recall, F1-score were used as performance measurements. All experiments performed on a CPU-based system.

4.2. Quantitative Evaluation

The performance of the text-only baseline with the proposed multimodal approach is compared in Table II.

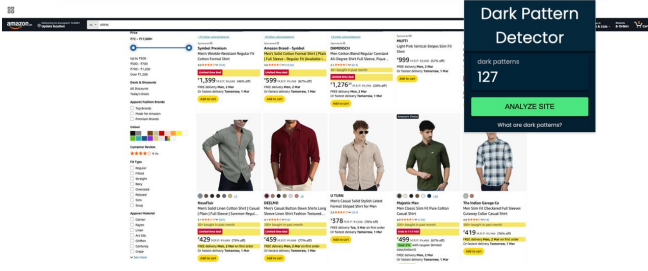


Fig. 2. Example output illustrating real-time detection of dark patterns on an e-commerce website.

Table 2. Performance Evaluation Results

Metric	Text-Only	Multimodal
Accuracy	87%	92%
Precision	0.85	0.91
Recall	0.83	0.90
F1-Score	0.84	0.90

The visual features provide a 5% increase in accuracy and significant improvement in recall. This demonstrates visual manipulability cues, such as urgency coloring and button prominence are not captured properly by text-based models.

4.3. Discussion

The conclusions of the research lend credence to the notion that dark pattern detection is inherently multimodal. Visual cues that are supplemented by them afford the advantage of furthering classification strength without incurring substantial computational cost. The system maintains real-time performance (≤ 500 ms processing time) but with 92% accuracy, showing fit for practical implementation on browsers.

5. CONCLUSION

This work proposed a lightweight multimodal framework for detecting dark patterns in web interfaces by combining textual classification with visual feature analysis. Compared to standard text-only techniques, our system captures persuasive language cues and visually driven manipulative signals directly from the DOM structure. Thanks to its modular browser-based design, it allows for real-time detection without the need for computationally intensive models or GPU resources.

Experimental observations reveal that the use of visual indicators enhances the overall detection accuracy from 87% to 92%, with remarkable improvements in recall. These results provide additional evidence for the notion that deceptive interface design functions by means of both linguistic framing and visual emphasis, and thus demands a multimodal view for accurate identification.

Although the framework reaches strong performance with minimal computational overhead, still several subtle or context-dependent manipulations are difficult. In future, we will work to increase the dataset, introduce adaptive feature weighting mechanisms, and conduct behavioral context modeling to further improve robustness and generalization across evolving web environments.

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